CNG 351 - Assignment 1

Data Management and File Structures

Fall 2026-2027

Entity-Relationship Database Designs

Team / Group ID: 8

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Notation and model organization

This report contains three conceptual database models. Each model is split into labelled drawing views for readability. Repeated entity names and keys in those views refer to the same entity, with attributes and relationships shared across the views.

Chen notation

Symbol / marking	Meaning
Single rectangle	Strong entity type.
Double rectangle	Weak entity; its full identity includes its owner's key.
Single / double diamond	Binary relationship / identifying relationship.
Oval / branched oval	Simple attribute / composite attribute with components.
Double oval / dashed oval	Multivalued attribute / derived attribute.
Solid underline	Chosen entity key. Other stated uniqueness constraints are listed in the rules.
Dashed underline	Partial key of a weak entity. BookingDate and StartTime together form Booking's partial key.
Single / double connection	Partial participation / total participation of that entity in the relationship.
(min,max) beside an entity	Number of relationship occurrences for each instance of that entity; N means many.

How to read a constraint

For example, Season (0,10) - Contains - Episode (1,1) means that each season contains at most ten episodes, while each episode belongs to exactly one season. The double line on Episode makes its mandatory participation explicit.

Views and additional rules

Use Case 1 uses Views 1A-1B; Use Case 2 uses Views 2A-2C; Use Case 3 uses Views 3A-3C. All relationship diamonds have exactly two entity roles. Relationship attributes do not add another entity role.

Temporal requirements, subset constraints and aggregate calculations that cannot be fully expressed by ordinary Chen symbols are stated as accompanying business rules. EmployeeCount* is derived automatically and also stored, as required.

Requirements source: CNG 351 Assignment 1, Fall 2026-2027, provided course handout. The eight vector drawing pages were exported from draw.io.

Use Case 1

Who Wants to Be a Millionaire?

Requirements: course assignment brief, pages 3-4.

The model separates the production of the show from events during a contestant's play. QuestionAttempt represents each asking of a question, while JokerUse records a particular use of a joker against one such attempt. This preserves the contestant, the exact question occurrence, the submitted answer and the joker usage time using binary relationships.

Entity identities

Entity	Chosen full key
Season	SeasonID
Episode	EpisodeID
Host	HostID
Contestant	ContestantID
Question	QuestionID
Category	CategoryID
QuestionAttempt (weak)	(ContestantID, AttemptNo)
JokerUse (weak)	(ContestantID, UseNo)
JokerType	TypeName

Design decisions

1. FullName is composite for Host and Contestant, with FirstName and Surname as components. Contestant.Age is derived from DateOfBirth using completed years.

2. Four fixed answer slots are shown as ChoiceA, ChoiceB, ChoiceC and ChoiceD. CorrectChoice identifies the correct slot. Category membership is a many-to-many relationship, with mandatory participation by Question.

3. QuestionAttempt is weak under Contestant. AttemptNo is a local sequence number, so it also preserves the order in which questions are asked. The link UsesQuestion identifies the question at that occurrence.

4. JokerUse is weak under Contestant and has one UsedFor link to a QuestionAttempt and one OfType link to JokerType. The contestant can therefore use zero to four jokers without creating a ternary relationship.

Use Case 1 - Rules and assumptions

Business rules and additional constraints

1. A season contains at most ten episodes; each episode belongs to exactly one season and is presented by exactly one host. A contestant belongs to exactly one episode. The system keeps one Contestant record for each person, enforcing a single participation in the show.

2. Host.Email is unique in addition to the chosen HostID key. EpisodeNumber is recorded as a separate attribute; the brief does not state that it is globally unique.

3. Each question has exactly four labelled answer slots, one CorrectChoice in {A, B, C, D}, question text and a DifficultyLevel label such as easy, medium or hard. Every question belongs to at least one category.

4. A completed contestant participation must have at least one QuestionAttempt. AttemptNo values are distinct within a contestant and increase in asking order. GivenAnswer is the submitted slot; the stored IsCorrect value must agree with GivenAnswer = Question.CorrectChoice.

5. The number of JokerUse occurrences owned by a contestant is between zero and four. Each occurrence has one date-time UsedAt, one joker type and one referenced QuestionAttempt. The attempt must be owned by the same contestant as the JokerUse.

6. The JokerType catalogue contains Fifty-Fifty, Phone a Friend and Ask the Audience, together with a description for each type.

Assumptions

A1. Seasons, planned episodes, hosts, categories and question-bank questions may be registered before they are used. Their unspecified minimum participation is therefore zero.

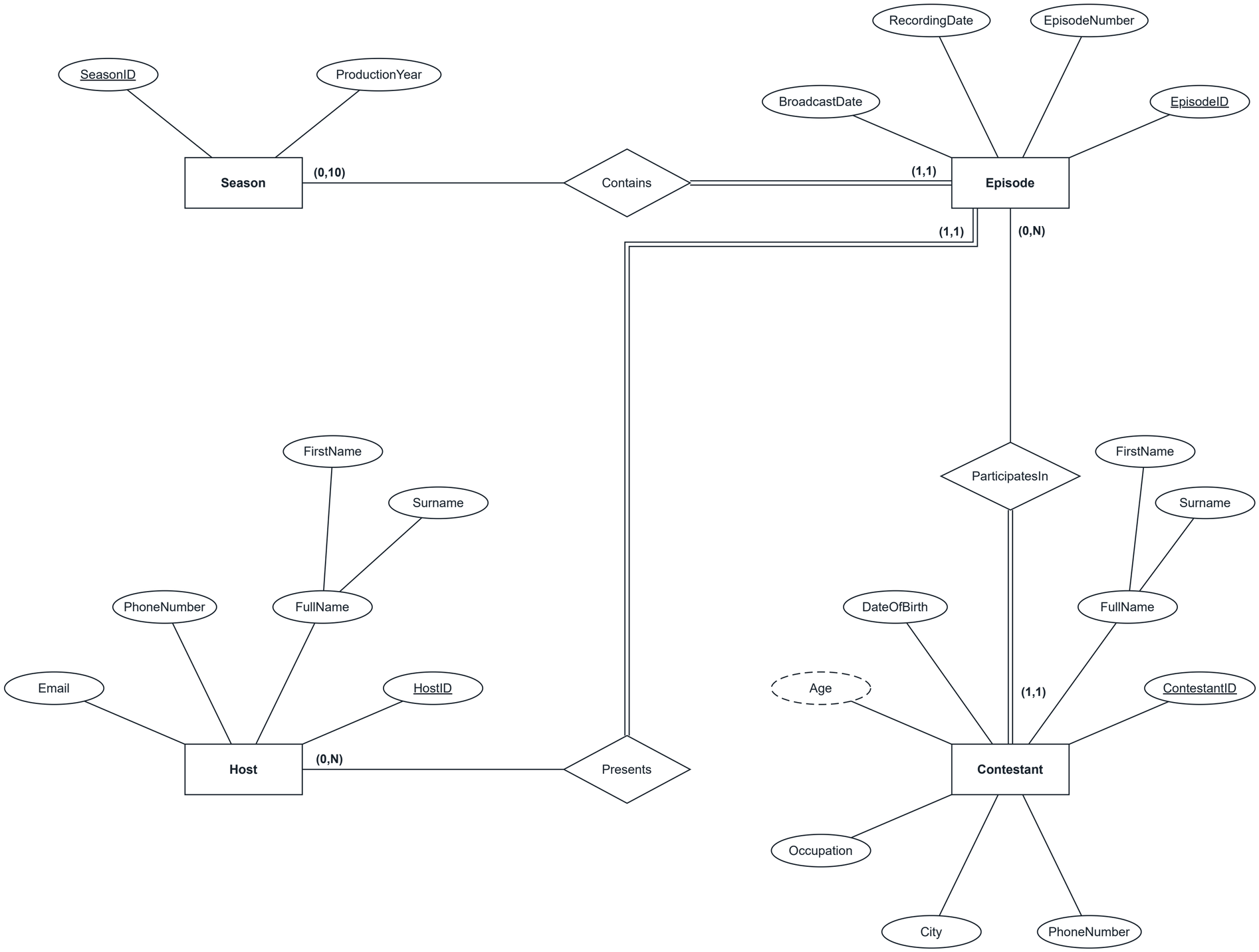
A2. The same question may be asked to a contestant more than once; separate AttemptNo values distinguish these events. No prohibition of repeated questions is stated.

A3. A joker type may be used more than once, subject to the stated total limit of four uses. The brief lists three types but gives no per-type usage limit.

A4. Every recorded contestant has been asked at least one question, so Contestant participates totally in HasAttempt.

Use Case 1A - Production and participation

Chen ERD | Entity attributes continue in View 1B where marked. All relationships are binary.

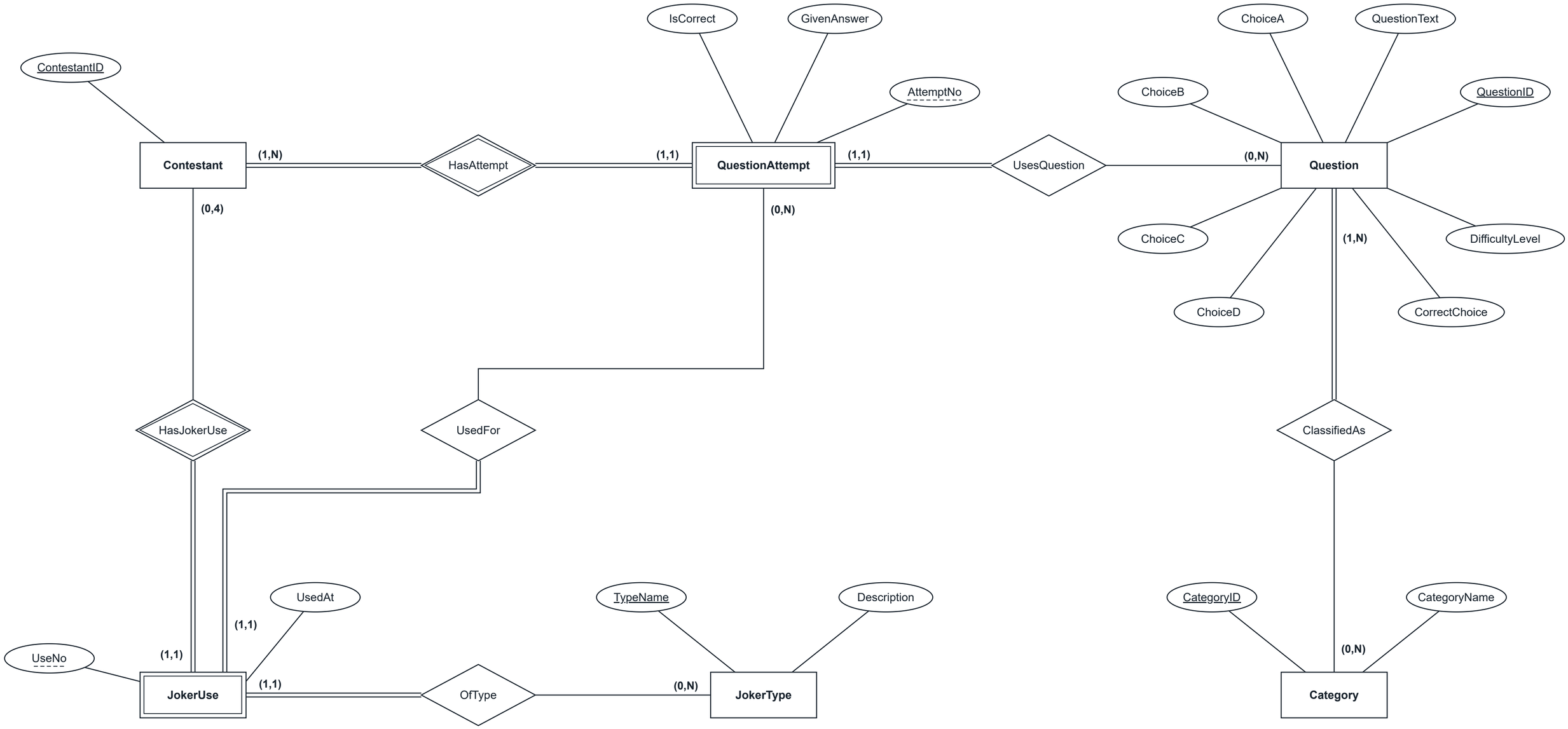


Age = completed years since DateOfBirth. Host.Email is an alternate unique key.

Contestant has one record per person and belongs to exactly one episode; therefore a person participates once.

Use Case 1B - Questions, answers and jokers

Chen ERD | Contestant is the same entity as in View 1A. Dashed underlines denote partial keys.



Full keys: QuestionAttempt = (ContestantID, AttemptNo); JokerUse = (ContestantID, UseNo).

A JokerUse must refer to an attempt owned by the same contestant. AttemptNo records question order.

TypeName domain: Fifty-Fifty, Phone a Friend, Ask the Audience. IsCorrect is recorded and checked against CorrectChoice.

Use Case 2

CleanCampus

Requirements: course assignment brief, pages 4-6.

MachineModel holds reusable equipment specifications. Machine-Dormitory allocation carries the rank and free credit for that particular pair. Current residence is recorded separately from individual Stay occurrences, allowing a student to return to the same dormitory. UsageSession is weak under Machine, and maintenance agreements are separate contract occurrences.

Entity identities

Entity	Chosen full key
LaundryRoom	RoomID
Machine	MachineID
MachineModel	ModelID
Dormitory	DormID
Student	StudentID
Stay	StayID
UsageSession (weak)	(MachineID, SessionNo)
Supplier	SupplierID
MaintenanceContract	ContractID

Design decisions

1. Each machine is based on one MachineModel. Capacity is composite, with MinLoad and MaxLoad, and SupportedPrograms is multivalued. Brand belongs to Machine as stated in the brief.

2. AllocatedTo is a binary many-to-many relationship. UsageRank and FreeCreditAmount belong to it. IsLeadingDorm is derived for each allocation from its UsageRank, rather than being entered independently.

3. Each period of residence has its own StayID and links to one Student and one Dormitory. CurrentlyIn records the student's current dormitory; the consistency rule below connects it to the open stay.

4. UsageSession has the partial key SessionNo and an identifying relationship to Machine. MaintenanceContract has its own identity, so multiple contracts between the same supplier and machine can be retained.

Use Case 2 - Rules and assumptions

Business rules and additional constraints

1. Machine.Type is standard or industrial. Dormitory.Level is undergraduate or graduate. Student.Username is unique in addition to StudentID. Student.Age is a stored attribute because no birth date is given.

2. Each machine is based on exactly one model and is located in exactly one laundry room. MinLoad must not exceed MaxLoad; the model records its supported program names.

3. Each Machine-Dormitory pair has at most one allocation, with its own UsageRank and FreeCreditAmount. IsLeadingDorm is true exactly when UsageRank = 1 for that machine.

4. Each student has exactly one CurrentlyIn dormitory and at least one recorded stay. Exactly one stay is current: its EndDate is NULL and its dormitory equals CurrentlyIn. A later return to the same dormitory creates a new StayID. Stay periods for one student must not overlap.

5. SessionNo is unique only within its machine. Status is scheduled, completed, cancelled or interrupted. Duration is calculated from StartTime and EndTime when both are known.

6. A completed or interrupted usage session must identify one dormitory. At the time a machine is actually used, that Machine-Dormitory allocation must exist. Scheduled or cancelled sessions may have no actual-use dormitory.

7. Every maintenance contract identifies exactly one supplier and one machine, and records StartDate, EndDate and ContractAmount. Supplier-machine renewals remain separate contracts.

Assumptions

A1. A machine is installed in one shared laundry room at a time. RoomID distinguishes these locations; historical room changes are not required.

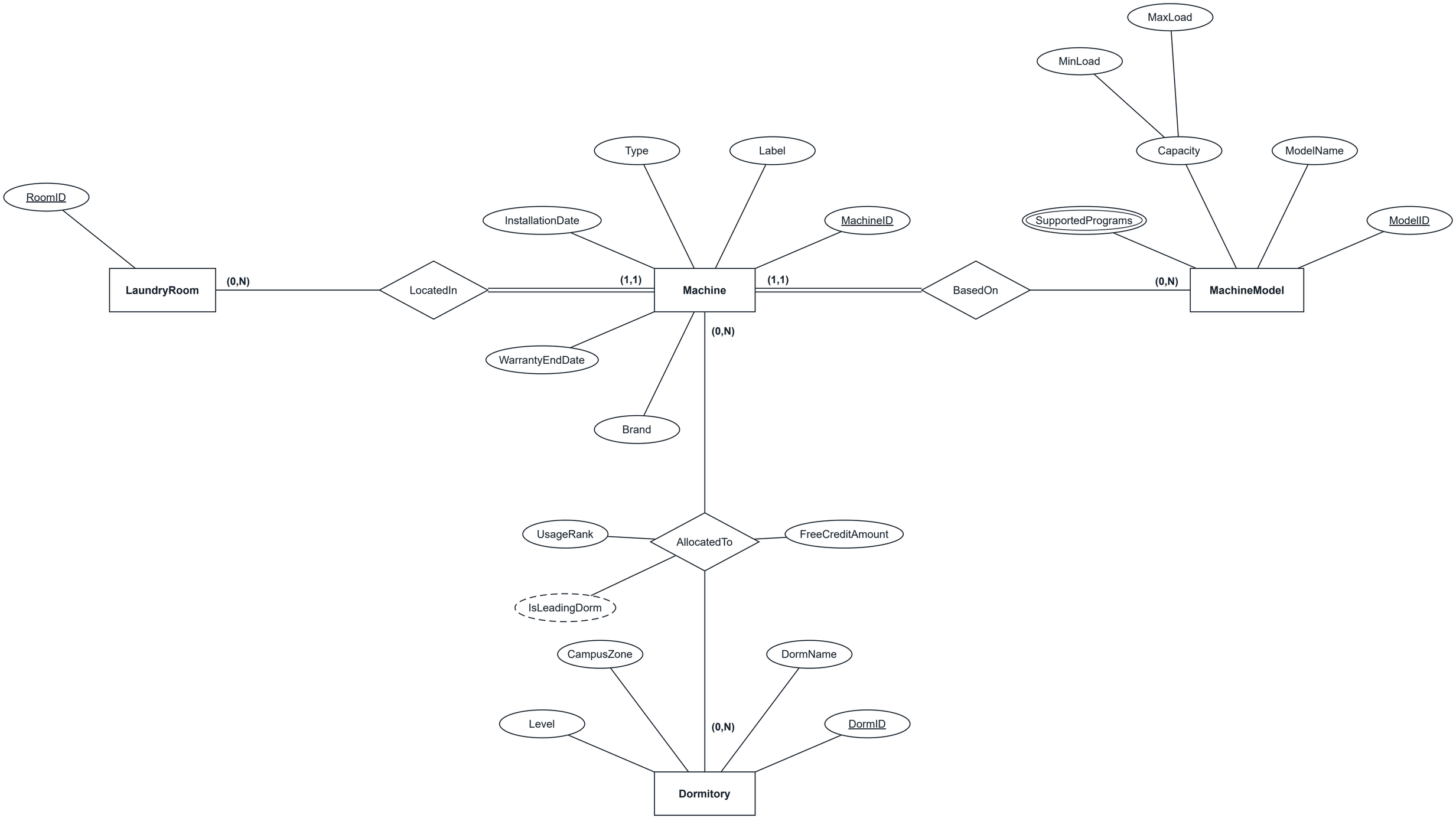
A2. Stay history includes the current open stay. EndDate is filled when a student leaves; CurrentlyIn is maintained consistently with that stay.

A3. Usage ranks are distinct positive integers within one machine's current allocations, with rank 1 identifying its sole leading dormitory.

A4. A usage session ends on its SessionDate. EndTime may be unknown for an unfinished or cancelled session; no duration is calculated until both time bounds are available.

Use Case 2A - Equipment and dormitory allocation

Chen ERD | Allocation attributes belong to the Machine-Dormitory pair.

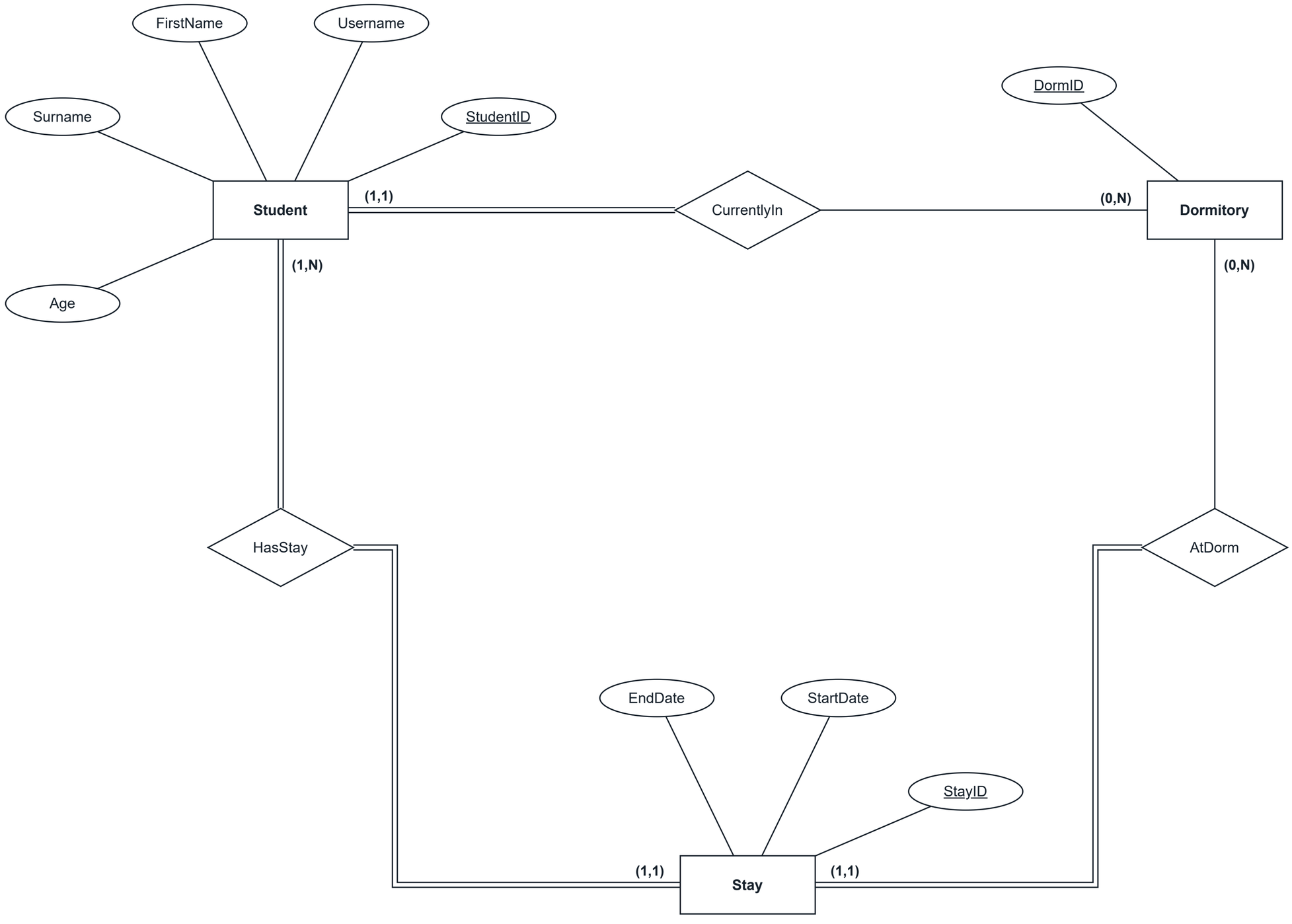


IsLeadingDorm = (UsageRank = 1), evaluated separately for each machine.

SupportedPrograms is multivalued. Capacity is composite: MinLoad and MaxLoad.

Use Case 2B - Current residence and stay history

Chen ERD | Dormitory is the same entity as in View 2A. Each separate stay has its own StayID.

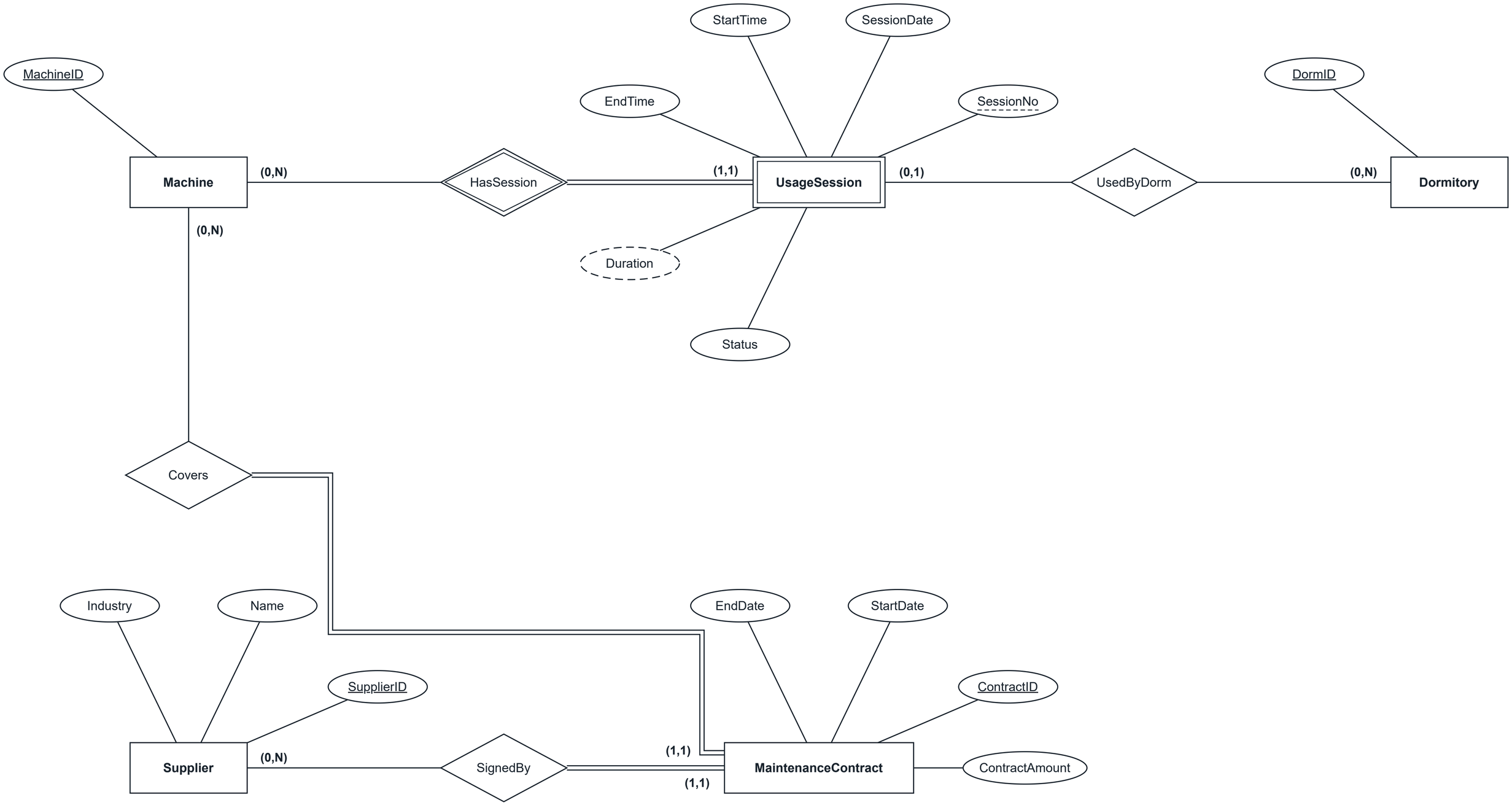


Student.Username is an alternate unique key. Student.Age is stored, as no birth date is supplied.

Exactly one current stay per student has EndDate = NULL and matches CurrentlyIn. A return stay gets a new StayID.

Use Case 2C - Usage sessions and maintenance

Chen ERD | Machine and Dormitory are references to View 2A. A usage session is weak under its machine.



Full session key: (MachineID, SessionNo). Duration is derived from StartTime and EndTime.

Completed/interrupted sessions require a dormitory, and the machine must be allocated to that dormitory.

A ContractID allows several contracts for the same supplier-machine pair across time.

Use Case 3

METU NCC KALTEV

Requirements: course assignment brief, pages 6-7.

The model records companies and their current employees, preserves office rental occurrences, and separates project deliverables, current internships, room bookings and investment transactions. Reified Rental and Investment entities preserve repeated events between the same parties. Cross-relationship rules enforce the representative, manager and supervisor requirements.

Entity identities

Entity	Chosen full key
Company	TaxNumber
Employee	NationalID
Office	OfficeNumber
Rental	RentalID
Project	ProjectCode
Deliverable (weak)	(ProjectCode, DeliverableNo)
Intern	StudentID
MeetingRoom	RoomNumber
Booking (weak)	(RoomNumber, BookingDate, StartTime)
Investor	InvestorID
Investment	InvestmentID

Design decisions

1. Company.Contact is composite, containing Email and PhoneNumber. Employee.PhoneNumbers is multivalued. CompanyAge is derived from FoundingDate.

2. EmployeeCount* is a materialized derived attribute: it is calculated automatically from current WorksFor links, stored for access, and refreshed whenever employment changes. The asterisk marks this storage requirement.

3. DeliverableNo is a partial key local to Project. Booking has the composite partial key (BookingDate, StartTime), local to MeetingRoom. Their identifying relationships preserve both ownership and the required uniqueness.

4. Investment represents one dated transaction. Company.TotalInvestmentReceived and investor-company totals are calculated from these transactions, so repeat investments are not overwritten.

Use Case 3 - Rules and assumptions

Business rules and additional constraints

1. TaxNumber and NationalID are the chosen Company and Employee keys. Employee.Email is also unique. Each employee WorksFor one current company; EmploymentStartDate is an attribute of that relationship.

2. Each company has one representative, and an employee represents at most one company. Represents must be a subset of WorksFor for the same employee-company pair. This also implies at least one employee per company.

3. Each Rental links one company to one office and records StartDate, EndDate and MonthlyRent. A rental is ongoing when StartDate is on or before today and EndDate is NULL or later than today. Every company currently operating in KALTEV has at least one ongoing rental. Ended rentals are kept as history, and an office may have no ongoing rental. Rental periods of the same office cannot overlap.

4. Each project belongs to one company and has status proposed, active, completed or cancelled. It has at least one deliverable. Deliverable numbering restarts at D1 for each project; ActualSubmissionDate may be NULL until submission.

5. WorksOn records HoursPerWeek for each employee-project pair. Each project has one manager. Manages must be a subset of WorksOn, so the chosen manager is a worker on that project.

6. Every intern InternsAt one company and has one supervisor. The supervisor must WorksFor that same company. Only the current internship is retained; the Intern record is removed when that internship ends.

7. There are exactly ten MeetingRoom records. Each booking has one room and one company. The full key (RoomNumber, BookingDate, StartTime) prevents two bookings with the same room, date and start time, while allowing simultaneous starts in different rooms.

8. Each Investment has one investor and one receiving company. Company.TotalInvestmentReceived is the sum of Amount over its investments, or zero when there are none. The total from investor i to company c is the sum over investments linked to both i and c.

Assumptions

A1. WorksFor records current employment; previous employer histories are not required. Employees work only on projects belonging to their current company.

A2. Rental periods are interpreted as half-open intervals [StartDate, EndDate), with NULL EndDate for an ongoing rental. A company may rent the same office again under a new RentalID.

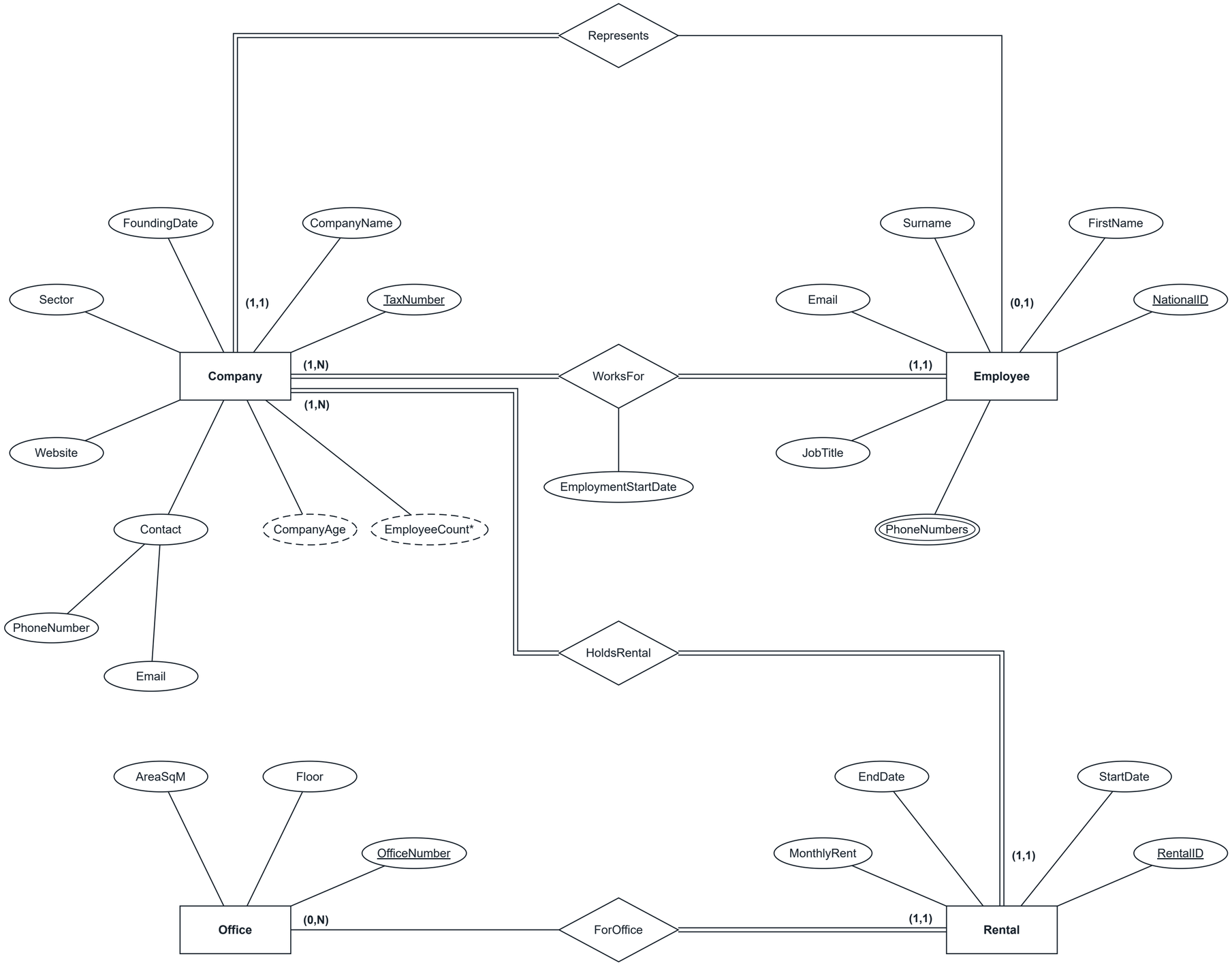
A3. A booking starts and ends on the same date, with StartTime earlier than EndTime. The key (RoomNumber, BookingDate, StartTime) prevents two bookings for the same room from starting on the same date at the same time.

A4. An investor may invest in the same company more than once. InvestmentID distinguishes those dated transactions. Investor email is not assumed to be unique. An investor may be registered before making any investment, so Investor participates partially in ProvidedBy.

A5. A company that has left KALTEV stays in the system so that its past rentals remain linked to it. The ongoing-rental rule applies only to companies currently operating in the building.

Use Case 3A - Companies, employees and offices

Chen ERD | Company.EmployeeCount* is a stored, automatically maintained derived value.



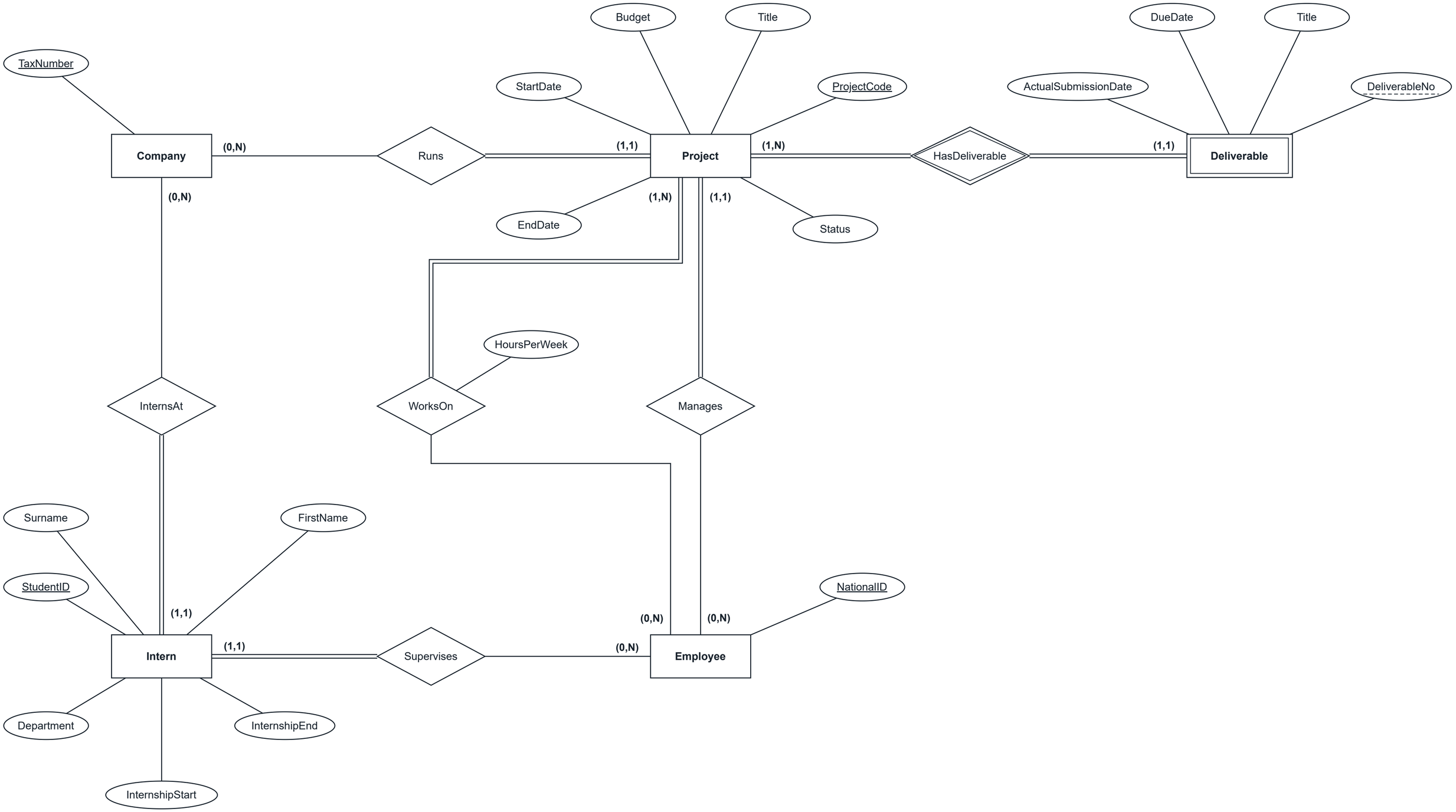
Represents must be a subset of WorksFor: a company representative is one of its own current employees.

Every company currently operating in KALTEV has at least one ongoing rental. Rental periods of the same office cannot overlap.

Employee.Email is an alternate unique key. EmployeeCount* = count of current employees in WorksFor.

Use Case 3B - Projects, deliverables and interns

Chen ERD | Company and Employee refer to View 3A. Every project manager must also work on that project.



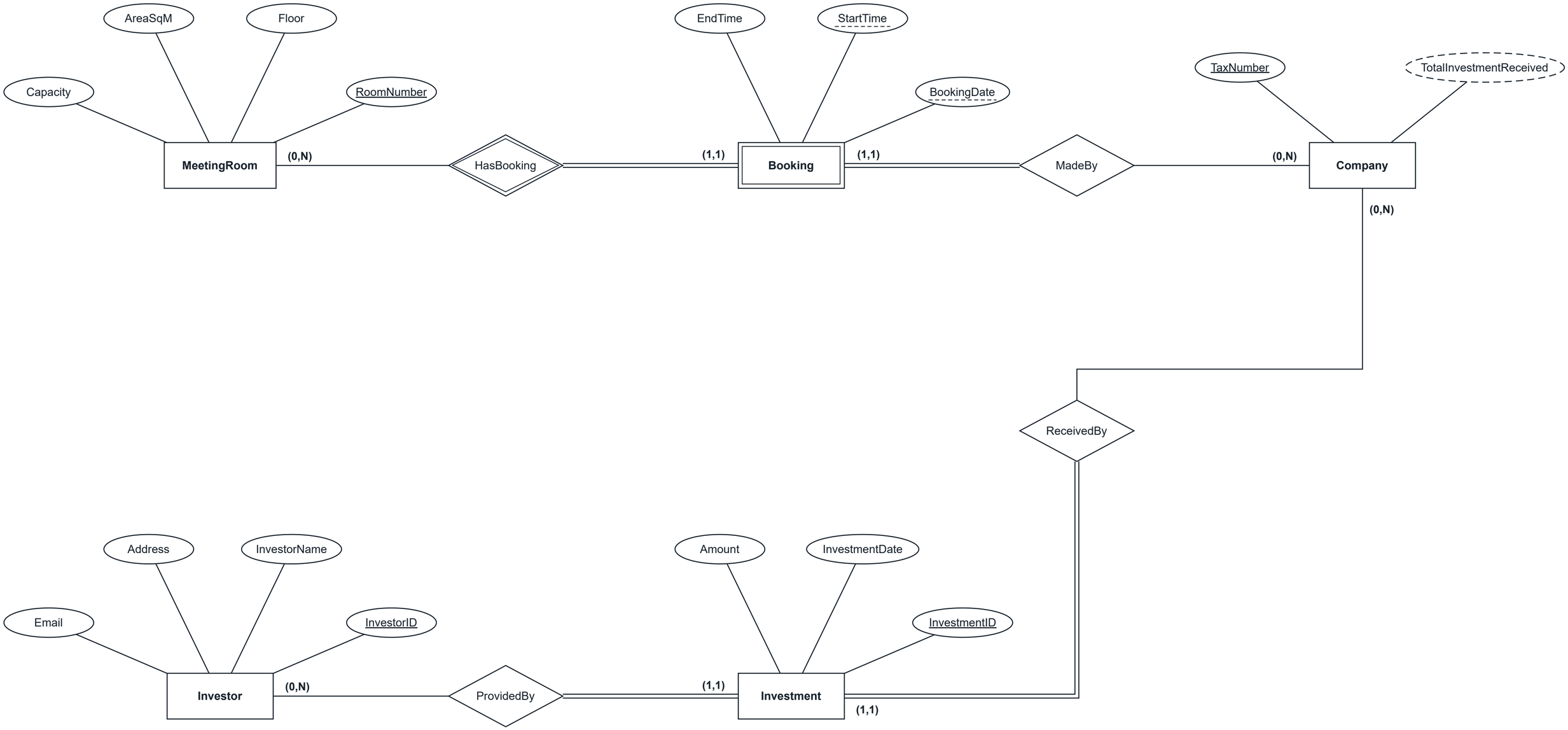
Full deliverable key: (ProjectCode, DeliverableNo). Numbering D1, D2, ... restarts within each project.

Manages is a subset of WorksOn. Project workers belong to the company that runs the project.

The supervisor must work for the intern's company. Only current internships are retained.

Use Case 3C - Meeting rooms and investment

Chen ERD | Company is the same entity as in View 3A; TotalInvestmentReceived is shown here.



Booking partial key is the pair (BookingDate, StartTime); full key also includes RoomNumber.

There are exactly 10 MeetingRoom records. A booking always belongs to one room and one company.

Investment totals are sums of Amount, by company or by investor-company pair. Repeated investments remain distinct.